# Badreddine Benfettouma

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## Research interests

Digital integrated-circuit design and RISC-V microarchitecture; formal verification of hardware; energy-efficient hardware for machine-learning inference; open-source EDA flows and open PDKs.

## Education

**Licence 3 (third year of the Bachelor), Electronics** 2026–2027
*Université Bourgogne Europe, Dijon, France*

**Licence 1 and 2, Computer Science and Electronics track** 2024–2026
*Université de Bourgogne (Université Bourgogne Europe since January 2025), Dijon, France*
Selected grades (out of 20): computer architecture 14.2 (lab 18.1), signal processing 17.5, scientific computing 20.

The French *Licence* is a three-year bachelor's degree (180 ECTS).

## Projects

**RV32I-Alpha: a two-stage RISC-V processor core** 2026
*VHDL, SystemVerilog; source code*

- Two-stage pipelined core implementing the 37 instructions of the RV32I base ISA, with traps on illegal instructions and misaligned accesses and a data-memory port that accepts wait states. Written in VHDL, then ported to SystemVerilog.
- Formal verification: riscv-formal instruction and consistency checks (bounded model checking, 10–15 cycles) and custom SymbiYosys properties for the memory ports; mutation testing (8 injected bugs, all detected); Yosys proof that the VHDL and SystemVerilog versions are sequentially equivalent.
- Implementation on a Digilent Basys 3 board (Artix-7, Vivado), and synthesis, placement and routing in SkyWater SKY130 with Yosys and OpenROAD (4 696 cells, 0.141 $mm^2$ die).

**Area and energy accounting of logic-gate neural networks** 2026
*Independent project, write-up in progress; Python, PyTorch, Yosys, OpenROAD, OpenSTA*

- Built an open flow that takes a trained logic-gate network to a placed SKY130 netlist, simulates it at gate level on real test images and derives power from the recorded switching activity.
- Checked that the gate-level netlist reproduces the software model bit for bit. Included the cost of input buffering, output logic and wiring, which published comparisons usually leave out.

**4-bit weight formats for LLM inference hardware** 2026
*Independent project; Python, PyTorch, Hugging Face Transformers; source code*

- Perplexity of five 4-bit weight formats on three LLMs. At equal quality, trellis coding saves 11–17 % of the bits against NVFP4 but only 6–9 % against a rotated Lloyd-Max format.

## Technical skills

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|---|---|
| HDLs | VHDL, SystemVerilog, Verilog |
| Verification | SymbiYosys, riscv-formal, Yosys equivalence checking, Icarus Verilog |
| Implementation | Vivado (Artix-7); Yosys, OpenROAD, OpenSTA, KLayout (SKY130 PDK) |
| Circuits | ngspice with SKY130 device models, Monte-Carlo mismatch analysis |
| Programming | Python (NumPy, PyTorch, Hugging Face Transformers), Tcl, RISC-V assembly; Linux, Git |

## Languages

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|---|---|
| French | native |
| English | C1 |
| Japanese | learning; level certification in progress |
| Arabic | Algerian Arabic (spoken); reads the Arabic script |

October 2026